

Title: A Fuzzy Congestion Control Protocol Based on Active Queue Management in Wireless Sensor Networks

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Presentation Outline

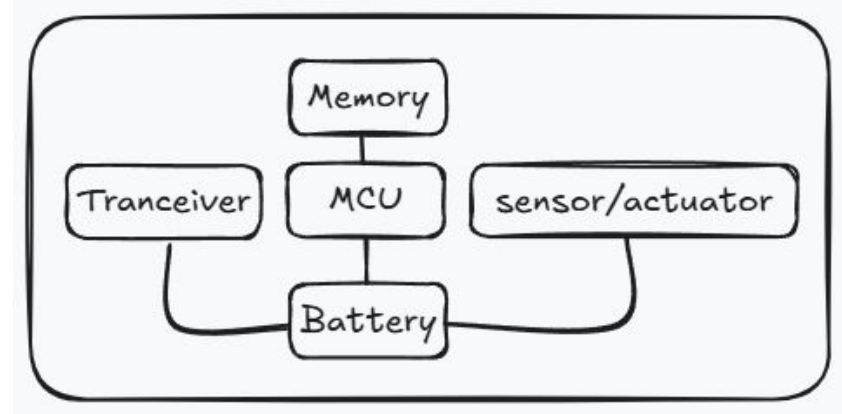
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Introduction & Background : Congestion and AQM in WSNs

A **Wireless Sensor Network (WSN)** is a collection of spatially distributed sensor nodes that monitor physical or environmental conditions (e.g., temperature, humidity, pressure, vibration).

Each node typically consists of:

- **Sensing unit** (to collect data)
- **Processing unit** (to process data locally)
- **Communication unit** (to transmit data)
- **Power unit** (usually battery-powered, resource-constrained)



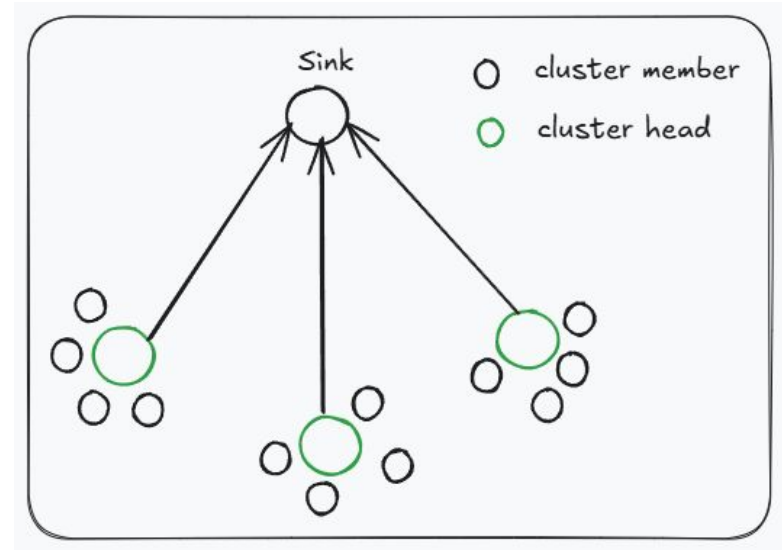
Routing in WSN

Routing in WSN differs from traditional networks due to **energy constraints, scalability issues, and dynamic topology**.

Common routing protocols:

- **Flat Routing** (e.g., Flooding, Gossiping) – all nodes play equal roles.
- **Hierarchical Routing** (e.g., LEACH, PEGASIS) – nodes are grouped into clusters, cluster heads manage communication.
- **Location-Based Routing** (e.g., GEAR, GPSR) – uses node positions to guide routing.

Efficient routing is essential to minimize **energy consumption and congestion**.



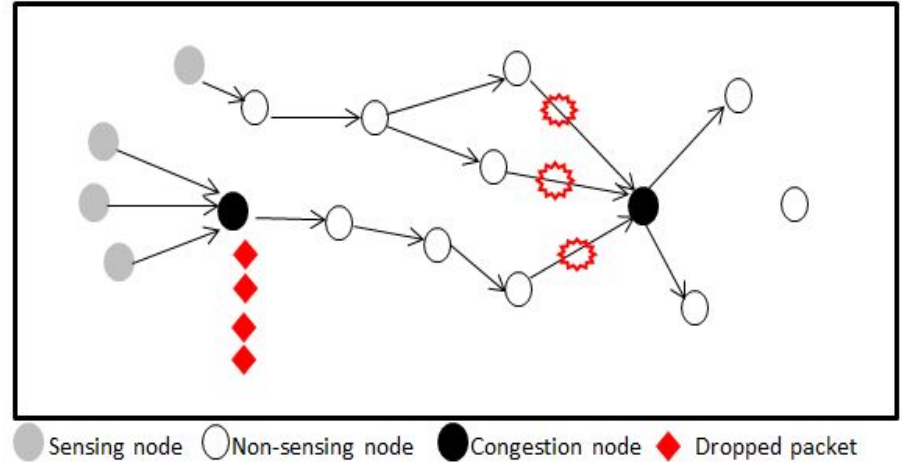
Low Energy Adaptive Clustering Hierarchy (LEACH)

1. Setup Phase - i) cluster head selection
ii) cluster formation
2. Steady state phase - i) data collection
ii) data aggregation
iii) data transmission to base station

How Does Congestion Occur in WSN?

Congestion in WSN happens when network resources (buffer, channel, energy) are **overloaded** due to:

- High traffic load from multiple sources.
- Burst transmissions during events.
- Limited buffer capacity in sensor nodes.
- Wireless interference and packet collisions.



Effects of congestion:

- **Packet loss, increased delay, wasted energy, and reduced network lifetime.**

Traditional Congestion Control

Inspired by **TCP-like approaches** in traditional networks.

Mechanisms:

- **Rate control** – nodes adjust data sending rates.
- **Queue management** – dropping or scheduling packets when buffer fills.
- **Backpressure routing** – rerouting traffic away from congested nodes.

Limitations in WSNs:

- TCP mechanisms are **not energy-aware**.
- Inefficient under dynamic, resource-constrained WSN conditions.

Fuzzy Congestion Control

Fuzzy Logic is used to handle **uncertainty and non-linearity** in WSN traffic.

Inputs (e.g., queue length, packet loss rate, link utilization) are mapped into **linguistic variables** (low, medium, high).

The fuzzy inference system determines **congestion level** and decides actions like **adjusting data rates** or **packet dropping priority**.

Advantages:

- Adaptive to varying traffic.
- Better resource utilization.
- Extends network lifetime.

Types of Fuzzy Congestion Control

Source-based Fuzzy Control: Adjusts sending rate at the source node.

Intermediate-node Fuzzy Control: Congestion handled at relay nodes by queue management.

Hybrid Fuzzy Control: Combination of both for better adaptability.

Fuzzy Congestion Control using Active Queue Management (AQM)

AQM is a queue management strategy that proactively manages buffer occupancy before it overflows.

In **Fuzzy AQM**, fuzzy logic is applied to monitor:

- **Queue length**
- **Packet arrival rate**
- **Packet loss probability**

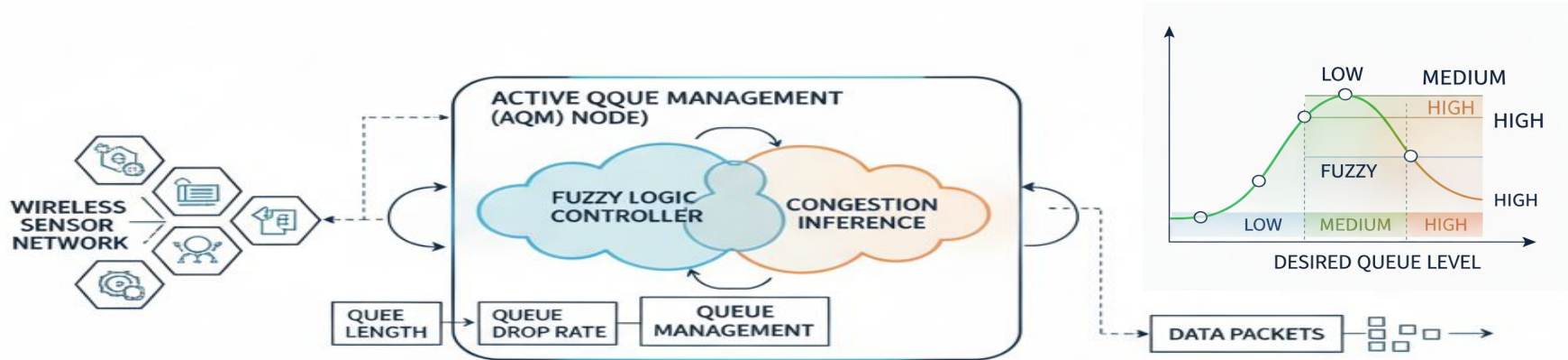
Based on fuzzy rules, the system decides whether to:

- Drop packets early.
- Adjust priority levels.
- Signal source nodes to reduce rates.

This approach reduces **packet loss, delay, and energy wastage**, improving **QoS and network lifetime**.

FUZZY CONGESTION CONTROL IN WSN

ACTIVE QUEUE MANAGEMENT



Comparison of Research Papers on Fuzzy Congestion Control in WSNs using AQM

Sl. No.	Paper (year)	Inputs / indicators used	Where fuzzy / AQM runs (node role)	Control action (output)	Eval / metrics	Key strengths	Key weaknesses
1	Rezaee & Pasandideh — (2017/2018).	Queue length (instant/avg), channel load, packet priority (medical data)	Cluster-heads / forwarding nodes (AQM at CHs)	Fuzzy → compute drop/mark probability + explicit CN to upstream; priority-aware dropping	Simulation (claims reduced packet loss & delay for medical traffic)	Application-aware (protects critical medical packets), proactive AQM.	Requires cluster formation; fuzzy engine overhead at CHs; parameter tuning needed.
2	P. Aimtongkham et al. — (2018).	Avg. queue length, arrival rate, packet loss; uses prediction + adaptive MFs	Node-level (distributed) — prediction + local fuzzy control	Fuzzy inference (adaptive MFs) → early drop / rate reduction; also predicts congestion to act earlier	Simulation (WSN scenarios) — throughput, delay, prediction accuracy	Adaptive MFs reduce manual tuning; prediction avoids late reaction.	Higher complexity; overhead of adaptive MF update; needs validation on real motes.

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3	FLACC (Fuzzy Logic Approach for Congestion Control) — (2019).	Average queue length (AQL), packet loss (PL)	Router / forwarding node (node-level AQM)	Fuzzy → compute drop probability (FLGRED variant) — early drop similar to RED but fuzzy-tuned	Simulation; compared to RED / GRED — delay & packet loss metrics	Simple 2-input fuzzy rule set; outperforms RED/GRED in heavy/bursty scenarios.	Simulation-only in original; energy/processing overhead not fully measured.
4	FBACCs (Fuzzy-based adaptive congestion control in WSNs) — (2013–2015)	Queue occupancy, arrival rate, sometimes channel load	Node-level / hop-by-hop	Fuzzy congestion estimator → smart early drop + regulate traffic rate	Simulation; compared with existing schemes — claims reduced packet loss & delay	Straightforward fuzzy estimator; rate regulation combined with dropping.	Multiple variants exist; some lack AQM explicitness; implementation details vary.

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5	A.Chakraborty — “Trust Based Fuzzy Algorithm for Congestion Control” (2013)	Queue length, trust metric of neighbour, packet loss	Node-level (hop-by-hop), trust-aware forwarding nodes	Fuzzy combines trust + congestion → rate adjustment and selective dropping	Simulation; metrics: throughput, normalized throughput, packet drop	Incorporates security/trust into congestion decisions (useful in WMSNs)	Extra overhead for trust computation; more state to maintain; not purely AQM.
6	L. Lin — “A Novel Fuzzy PID Congestion Control Model” (2020)	Instantaneous queue length, packet loss rate (real-time)	Node-level / gateway nodes (control loop at forwarding nodes)	Fuzzy-PID → produces control signal for rate adjustment (and early-drop decisions)	Simulation (NS-style): queue delay, packet loss, stability	Combines control-theory stability (PID) with fuzzy tolerance to noise → smoother control.	More complex controller; tuning PID + fuzzy parameters required; overhead not profiled on motes.

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7	AA Abu-Shareha et al. — “Fuzzy-Based Active Queue Management Using Precise” (2023)	Queue metrics (avg qlen), traffic descriptors	Router / AQM-enabled node; methodology portable to WSN gateways	Fuzzy AQM → compute drop/mark probability; optimizes fuzzy model with GA	Simulations (network-level) — packet loss, delay improvements	Efficient fuzzy modeling + GA optimization → better AQM tuning.	Paper is general network AQM (not WSN-specific) — needs adaptation for resource-constrained nodes.
8	N. Mazloomi et al. — “Efficient fuzzy methodology for congestion control” (2024)	Queue, link status, possibly prediction outputs (SVR)	Cluster heads / gateway (hybrid for larger deployments)	Fuzzy + ML (SVR) → AQM-style decisions and rate control; hybrid approach	Simulation/case studies — energy, packet loss, stability	Hybrid fuzzy + ML improves accuracy in complex networks.	More computation due to ML components; complexity & energy cost tradeoff.

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9	Sonmez et al. -“Fuzzy-based congestion control for Wireless Multimedia Sensor Networks” (2014)	Queue length, delay, frame importance / application priority	Gateways / cluster heads (AQM at aggregation points)	Fuzzy → prioritize important frames (I-frames) and compute drop probabilities for less important frames	Multimedia metrics (PSNR, frame loss), delay	Cross-layer design that preserves multimedia QoS in congestion.	Designed for WMSN (large packet sizes); not general-purpose sensor networks.
10	Hatamian et al. — congestion-aware routing + fuzzy rate controller (2015–2017)	Local load (queue), neighbor states, link quality	Routing nodes (distributed) — fuzzy used inside routing/forwarding decision	Fuzzy produces next-hop weight + rate-adjustment; AQM-like early drops when necessary	Simulation: throughput, packet drop, energy lifetime	Avoids hotspots by combining routing + fuzzy congestion control.	Requires state exchange overhead; more complex to implement.

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11	IFCCDC (IF Fuzzy Control based Congestion Detection & Control) — (2012).	Queue occupancy, channel conditions	Node-level (local detection & control)	Fuzzy detection → local drop / notify upstream; simple fuzzy rules	Simulation/academic tests — delay, drop reduction	Early fuzzy detection reduces packet loss; straightforward to implement.	Older/short conference paper; limited evaluation detail; not AQM-optimized.
12	Rajeswari et al. — “Fuzzy based congestion detection and control algorithm” (2022)	Queue length, delay, packet loss; mentions type-2 fuzzy variants	Node-level / cluster-heads	Type-2 fuzzy estimation → early-drop / rate regulation; priority-aware options	Simulation; QoS and lifetime metrics	Type-2 fuzzy gives better handling of uncertainty in noisy WSN conditions.	Type-2 fuzzy inference is heavier computationally; practical overhead needs measurement.

Motivation for the Study

Rapid growth of WSN applications

- Used in healthcare, military, environmental monitoring, and smart cities.
- Require reliable and real-time data delivery.

Challenge of congestion in WSNs

- Limited bandwidth, buffer size, and energy resources.
- Congestion leads to **packet loss, high delay, energy drain, and reduced network lifetime**.

Limitations of traditional congestion control

- Rely on fixed thresholds and rigid queue management.
- Lack adaptability to dynamic traffic and varying node conditions.
- May not balance trade-offs between throughput, delay, and energy efficiency.

Why Fuzzy Logic?

- Handles **uncertainty and imprecision** in congestion indicators (queue length, packet delay, packet loss).
- Provides **adaptive decision-making** instead of binary threshold-based control.

Role of Active Queue Management (AQM)

- Prevents congestion proactively rather than reactively.
- Early detection and control of queue buildup to maintain **smooth data flow**.

Research Gap

- Few protocols integrate **fuzzy inference with AQM** in WSNs.
- Need a solution that is **lightweight, adaptive, and energy-efficient** while ensuring **QoS (Quality of Service)**.

Experiment Plan

1. **Implement baseline:** simple Drop-tail + CSMA, and RED.
2. **Implement fuzzy AQM variant:** choose 3 inputs (AQL, packet loss, priority).
3. **Comparator protocols:** FLACC, a priority-aware fuzzy scheme (PCCF variant), and one fuzzy-AQM paper implementation (e.g., 2017 medical WSN paper logic).
4. **Simulator:** NS-3 or OMNeT++ with realistic traffic (periodic + event bursts). Evaluate: packet delivery ratio (PDR), end-to-end delay, energy consumed, and fairness.
5. **Measure overhead:** simulate/estimate CPU cycles for fuzzy inference or implement on a mote to measure energy/time.

Novelty of Our Project

Aspect	Existing Work	Our Work (Novelty)
Baseline Protocols	Usually compare with only one baseline (e.g., DropTail or fuzzy variant).	Compare with DropTail + CSMA and RED → stronger fairness baseline.
Fuzzy Inputs	Typically 1–2 (e.g., queue length, delay).	3 inputs : AQL, Packet Loss, Priority → QoS-aware congestion control .
Protocol Comparison	Limited (often a single fuzzy scheme).	Benchmark against FLACC , PCCF variant , and 2017 fuzzy AQM → broader evaluation.
Simulation Environment	MATLAB or simplified simulators.	NS-3/OMNeT++ → realistic MAC, collisions, retransmissions, energy models.
Performance Metrics	Focus mainly on throughput/delay.	Evaluate PDR , delay , energy , fairness → more comprehensive.
Overhead Analysis	Assumed fuzzy is lightweight, rarely measured.	Explicitly measure CPU cycles/energy for fuzzy inference → shows practical deployability.

Conclusion

- **Problem Addressed**

- Congestion in WSN leads to **packet loss, delay, and energy wastage**.
- Traditional DropTail and RED are too rigid, and existing fuzzy schemes are narrow.

- **Our Contribution**

- Proposed a **Fuzzy AQM protocol** with **3 inputs (AQL, loss, priority)**.
- Benchmarked against **DropTail + CSMA, RED, FLACC, PCCF, and a fuzzy AQM scheme**.
- Used **realistic simulation (NS-3/OMNeT++)** with periodic + burst traffic.

- **Findings**

- Achieves **higher PDR, lower delay, and better fairness**.
- Reduces **energy waste** under congestion.
- Overhead analysis shows **fuzzy inference is lightweight** and deployable.

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